

BSc (Hons) Computing - Coventry University
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Data Warehousing and Business Intelligence

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6.1.A Data Concepts

Our Objectives



What we are going to learn?

- Define and differentiate between data, information, and knowledge.
- Recognize different types and dimensions of data.
- Understand the concepts of big data and metadata.
- Learn the basics of data storage, processing, ETL, and data mining.
- Recognize the need of data management for efficient data retrieval.

What is Data?

- **Definition of Data: Raw facts and figures**

- **Quick brainstorming: “What types of data do you encounter daily?”**

Examples

- Sales numbers from a local supermarket
- Temperature readings in Colombo

Quantitative vs. Qualitative Data

- My best friend stands at 5 feet 7 inches tall.
- They wear a size 6 shoe.
- Their weight is 63 kilograms.
- They have one older sibling and two younger ones.
- They own two cats.
- My best friend lives about twenty kilometers from me.

- My best friend has curly hair.
- Their eyes are black.
- Have a great sense of humor, are loud, and are a fantastic listener.
- Sometimes, they can be a bit impatient and impulsive.
- My friend has a friendly face and a laugh that's contagious.

*Credit:
careerfoundry.com*

Data, Information, and Knowledge

- **Data → Information:** Data processed to provide meaning
- **Information → Knowledge:** Understanding gained from information

Data is like *ingredients*, information is *a recipe*, and knowledge is *understanding how to improve the dish*.

Discussion

How do we use data, information, and knowledge in everyday life?

- **Data:** Locations on a map.
- **Information:** The directions and routes between those locations.
- **Knowledge:** The traveler's insights on the best times to visit, local customs, and hidden gems along the way.

Types of Data



- Structured: Organized, easily searchable data (e.g., spreadsheets, relational databases).
- Unstructured: No predefined format (e.g., images, videos, social media).
- Semi-Structured: Some structure but flexible (e.g., emails, JSON files).

Example:

A supermarket like Cargills Food City may have sales data (structured) but customer feedback in emails (semi-structured).

Discussion

The difference - Spreadsheets vs. Databases?

- **Spreadsheets:** Suitable for small data sets, personal analysis.
- **Databases:** Optimized for large volumes, multi-user environments.
- **Limitations of Spreadsheets:** Limited data capacity, lack of advanced querying.
- **Example:** A bank might use a database rather than a spreadsheet to store customer information.

Dimensions of Data

- **Time: Historical vs. real-time data**
- **Subject: Different categories (e.g., sales, inventory)**
- **Location: Geographic context (e.g., regional sales data)**

An Internet Minute [2023]



Credit: Domo

https://en.wikipedia.org/wiki/Internet_minute

Big Data

Big data refers to data sets that are too large or complex to manage with traditional tools.

Big data is like a vast ocean - navigating it requires powerful tools.

Example

Analyzing social media trends during events, like elections or festivals.

Challenges

Storage, processing, and deriving insights.



Fundamental Issues with Big Data

- ***Data Quality: Inaccuracies and inconsistencies***
- ***Data Integration: Combining data from various sources***
- ***Privacy Concerns: Protecting personal information***



Volume, Variety, Velocity, Veracity

Data Quality Dimensions (DAMA UK)

2. Completeness: The proportion of stored data compared to the total data that should be available.

- *Example: A healthcare database with all necessary patient records and medical history details.*

1. Accuracy: How well the data represents real-world facts or events.

- *Example: Ensuring a customer's address is accurately recorded in the records of a delivery service.*

3. Uniqueness: Whether the data is stored in multiple places without unnecessary duplication.

- *Example: A national ID number that is unique and appears only once in government databases.*

Data Quality Dimensions (DAMA UK)

Continued...

5. Timeliness: The degree to which data is up-to-date and available when required.

- *Example: Real-time updates on public transportation schedules to ensure commuter reliability.*

4. Consistency: The absence of conflict between data from multiple sources.

- *Example: Financial reports from different branches of a company should not show conflicting revenue figures.*

6. Validity: The extent to which data conforms to defined business rules or formats.

- *Example: Customer phone numbers must follow a valid format in a telecommunications provider's database.*

Metadata



Data about data: Helps in data management and understanding context

Metadata is data about data, providing context and meaning.

- **Purpose:** Helps in organizing, finding, and understanding data.
- **Example:** Metadata in a library system - catalog details help users find books.
- **Analogy:** Metadata is like an index in a book, guiding us to the right information.

Discussion

Think of a real-life example of metadata in a university setting.

Data Storage and Processing Techniques

- **Data Storage Options: Local, cloud-based, distributed storage.**
- **Processing: Batch (large data sets at once) vs. real-time (instant processing).**

Introduction to Data Management

Organizing and maintaining data to meet information needs.

- **Importance:** Ensures data quality, security, and accessibility.
- **Analogy:** Data management is like maintaining a library - orderly, secure, and easy to access.

What might happen if data isn't managed properly?

Facilitating Data Retrieval

- **Key Requirements: Organized storage, efficient indexing, and metadata use.**
- **Challenges: Large volumes, varied formats, data accuracy.**
- **Example: In banking, rapid access to customer transaction history is essential.**

Questions and Discussion

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Last modified: 04 Nov 2024